# Video Transcript

## Video 2: Understanding K-Means (07:49)

So, let's take a look now at an algorithm called K-Means that allows us to cluster blobs of data together. So, this is built into sklearn, but we're going to write our own algorithm, so that we can understand how it works. Now the first thing is that we can use a generator to make blobs of data. So here, we're going to make a number of clusters. Here, I'm making two clusters of data. So, we can change this number and make three clusters, four clusters, etc. Let me just show you. So, let's suppose we wanted four clusters. We would have four clusters. Let's go back to two clusters, and you can work through it with a different number. So, here's our two clusters of data. We can plot them out. So, this is saying we'll plot all the X's, and that's the first value, and this is the second value. So, if you like, it's an X, Y. So, we're just plotting X against Y here, and we've got 300 points.

So, let's take a look at how the algorithm works. So, now we call it and we say, okay, I want to divide this. I know I generated only two clusters, but I want to generate it into four. So, K-Means, we can just choose, and we have to choose how many clusters. It doesn't tell us automatically. Let's leave it at four here for now. So, it's dividing it into four clusters, and we can plot out the centers of those clusters. It turns out that the algorithm requires us to calculate the center, and what we mean by the center is we give each point the same weight and calculate where the centroid is of that cluster. So, we start off with four arbitrary points, and we cluster around those points and those points actually move. Let's take a look at an algorithm that we've written ourselves. So, this would be if we didn't want to use the sklearn one, we could write our own. So, what we're going to do is we're passing in the X, Y coordinates of all the points, and we say, how many clusters we're going to have. Now, we generate from all of the X readings. We generate a number of clusters. So, here we're going to call the algorithm eventually with a number of clusters. So, here find_clusters and we're passing in four. This rseed is just a seed that we use here. But when we see this, what we're going to get here for this i, let's print it out, we'll see that we get four points. So, we've got a whole string of 300 points, and we're going to choose the center where we're going to start for each cluster as whatever the coordinates of that 84th point are, the 217th, the 211th, and the 286th. So, these are arbitrary starting points

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for our centers, and in this case, for four clusters. We're going to keep the history of where those centers are because those centers are going to move. So, first we find the distance between all the points and the centers, the pairwise distances.

So, let me show you now, how we go about using the pairwise distance comparison to update the centers. Here, I've got two centers chosen at random, the blue one and the green one. Now, we calculate the pairwise distance of all the points, and you can see here that it's pretty obvious some of the points belong to the blue and some to the green, and a couple of them are quite close. So, what we do now is label the points according to which center they're closest to. So, here we've got the green ones and here the blue ones. Now, we update the center again. So, you can see the green centroid is not in the right place at the moment, it needs to be updated. So, we update it, and now what we see is that another point, this point here has become closer to the green center than to the blue center. So, this one will get updated and will be re-labeled as green and then again, the centroids will be moved. At some stage, no points are relabeled, and so everything is static and that's the end of the labelling algorithm. And so, the clusters then are well-defined, okay? So, we do this and then we calculate the centroid. So, what we're going to do here is we're going to calculate where the new centers are. Once, we've assigned the labels, we're going to calculate now the centroid of each of those labels. So, that'll be, and it'll move. And we're going to record now where it's moved to, in this history,

and we go around again now and calculate the new distances of each point to those new centers. So, let's do it, and we'll see that we get a plot here, and what you can see is you can see the centroids moving. This green center started here, and then moved to the right and then came back a little. The yellow one looks like it started here, and moved a little to the left. You can see up here, this one moved from the left here towards the right for that cluster. And then, this one moved a little in that cluster. We stop our loop when we get no change of labels on points, when all the points stay statically attached to a given label, in this case a given color. It's a very simple algorithm, but quite powerful. So, why don't you play with this? So, for example, we can change the number of centers that we're going to find. So, let's see what would happen if we chose two centers. So, here we'd break up and you can see the centroid moving. So, in this case, it actually probably moved from the left here. It looks like this would be the centroid of these points. Okay, now suppose some of the points had color already assigned to them. How would that change our algorithm? So, now we've got points that are close, but they may be a

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different color. Well, it depends, because it's a little bit like comparing apples and oranges. In one case, we've got a distance, and in another case, we've got a color. So, unless we have some means of saying how we trade-off those metrics, it's difficult for us to choose how to find that distance.

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